30-Day Technical Training Roadmap

Goal: Prepare Onshore + Offshore Team for Retail Data Engineering Projects (AWS + Snowflake + dbt)

WEEK 1 — Foundations & Environment Setup

Onshore (You):
• Set up AWS accounts, IAM roles, S3 structure.
• Create Snowflake account, warehouses, roles.
• Install dbt Core & configure profiles.yml.
• Learn Snowpipe fundamentals.
• Create canonical retail schema (sales, product, store, customer).

Offshore Team:
• Install AWS CLI, boto3, Python dependencies.
• Learn Lambda basics (handlers, triggers, retries).
• Learn Git fundamentals (branches, commits, pull requests).
• Learn S3 folder structure & raw zone design.
• Write first sample ingestion script (dummy API → S3).

Deliverables:
✓ AWS + Snowflake + dbt fully configured
✓ Dummy API ingestion → S3 pipeline
✓ GitHub repo created with branching rules

WEEK 2 — Ingestion & Snowflake Setup

Onshore (You):
• Configure Snowpipe for automatic S3 ingestion.
• Create staging schemas for client1 (client1_raw, client1_stg).
• Build example dbt staging models (stg_sales, stg_products).
• Define data quality rules (unique, not_null, referential).

Offshore Team:
• Build 2 real ingestion jobs (CSV & API).
• Implement CloudWatch logging & error alerts.
• Write Glue crawler to infer schema on raw files.
• Learn incremental ingestion patterns.

Deliverables:
✓ Snowpipe loading data into Snowflake
✓ Working dbt staging models + YAML tests
✓ 2 ingestion jobs completed and logged

WEEK 3 — Transformations & Marts

Onshore (You):
• Design full canonical model (fact_sales, dim_product, dim_store).
• Create dbt intermediate + core layer structure.
• Build incremental fact_sales model.
• Review offshore code and implement CI/CD for dbt (GitHub Actions).

Offshore Team:
• Build dim_product, dim_store models.
• Write dbt tests (unique, not_null, foreign key).
• Configure dbt docs + generate documentation site.
• Create S3 → Snowflake backup scripts.

Deliverables:
✓ Full dbt project skeleton (stg/int/core/marts)
✓ Incremental fact table working
✓ CI/CD pipeline for dbt

WEEK 4 — Dashboards, Monitoring & Final Integration

Onshore (You):
• Define KPIs for retail dashboards (sales, inventory, staff performance).
• Connect Power BI/Tableau to Snowflake.
• Create final dashboard templates.
• Conduct code reviews & data validation.

Offshore Team:
• Build dashboard visuals based on your template.
• Implement CloudWatch alarms for pipeline failures.
• Create S3 archival lifecycle rules.
• Finalise dbt docs & QA checks.

Deliverables:
✓ Fully automated pipeline (API → S3 → Snowpipe → dbt → Dashboard)

✓ Power BI/Tableau retail dashboards ready for demos
✓ Monitoring, alerts, documentation complete

After Day 30 — You are ready for real client projects.

Your team will be capable of delivering small-to-mid scale retail analytics projects, including ingestion, modelling, and dashboard delivery. This foundation is enough to onboard your first 1–3 paying clients and demonstrate value quickly.